

Assessing Human-LLM Alignment in the Evaluation of Odia Spelling and Grammatical Errors

HSL7702: COMPUTATIONAL LINGUISTICS (NATURAL LANGUAGE UNDERSTANDING)

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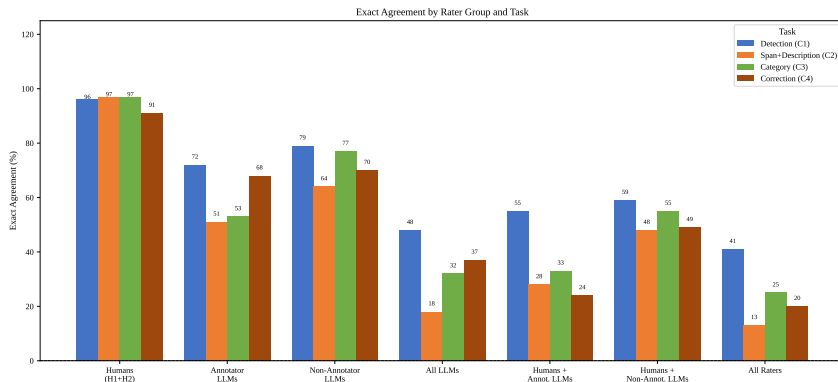
Submitted to: **Prof. Ashwini Vaidya**

May, 2026



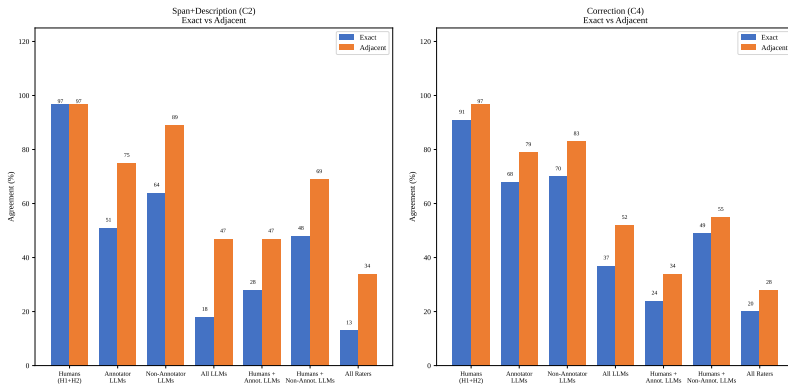
Appendix: All Figures and Tables

A1. Exact Agreement by Rater Group and Task



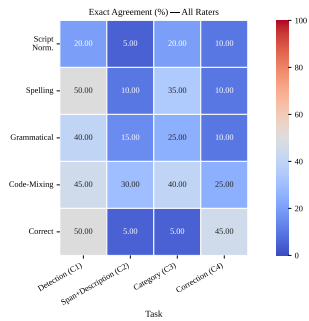
Humans (G1) achieve 91 to 97% exact agreement across all tasks. The annotator LLM pair (G2) reaches only 51 to 72%, the lowest of any two-rater group. As group size grows toward G7, unanimous agreement drops to 41% on C1 and 13% on C2, expected given six diverse systems.

A2. Exact vs. Adjacent Agreement for C2 and C4



For G1, the gap between exact and adjacent agreement is negligible (e.g. C4: 91% vs. 97%), meaning human disagreements are always just one scale step apart. For G2 the gap is larger (C2: 51% vs. 75%), but this still shows that Sarvam and LLaMA rarely disagree by the maximum 0-to-2 distance.

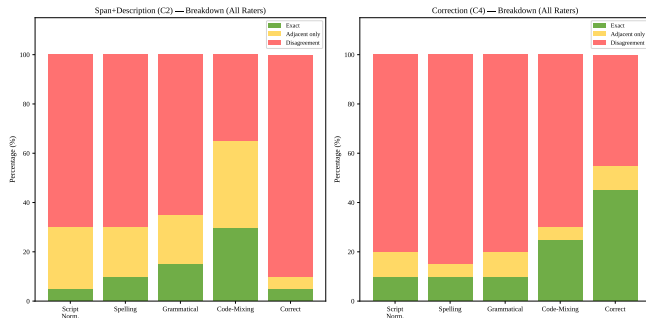
A3. Exact Agreement Heatmap by Error Category (All Raters, G7)



Code-Mixing / Wrong Language yields the highest cross-rater exact agreement on C1, consistent with the visual distinctiveness of Roman-script words in an Odia sentence. Grammatical Error shows the lowest C4 agreement, reflecting the inherent ambiguity in what counts as an acceptable Odia correction.

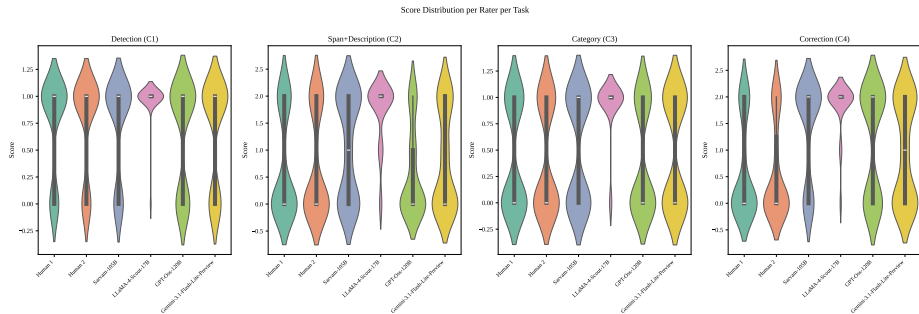


A4. Stacked Agreement Breakdown for C2 and C4 (All Raters, G7)



The “full disagreement” bar (scores more than one step apart) is small for most categories, confirming that even when evaluators disagree on span quality or correction quality, extreme disagreements are rare. Grammatical Error has the largest full-disagreement component on both C2 and C4.

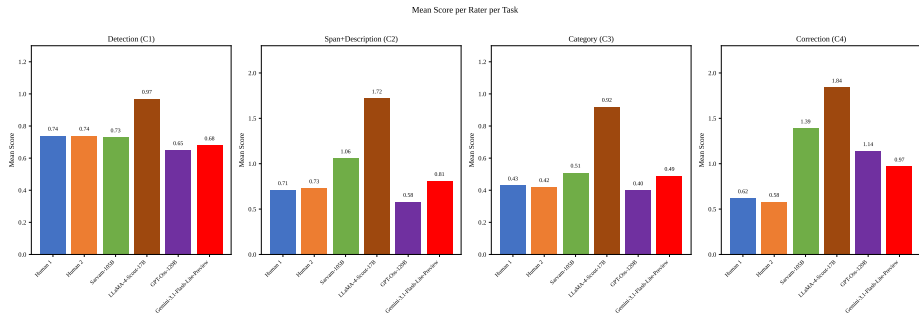
A5. Score Distribution per Rater per Task



LLaMA assigns 2s on C2 and C4 far more often than any other rater, producing a distribution skewed toward the top of the scale.

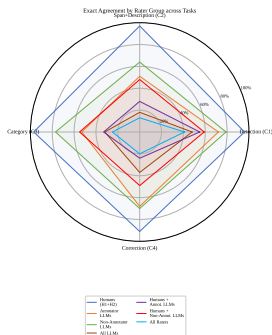
Human 1 and Human 2 show tightly matched distributions across all tasks, visually confirming their near-perfect agreement.

A6. Mean Score per Rater per Task



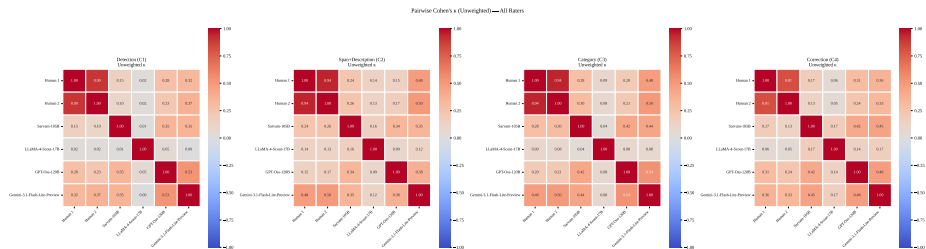
LLaMA's mean scores on C2 (1.72) and C4 (1.84) are substantially higher than those of all other raters, which cluster between 0.8 and 1.1. This leniency bias is the primary driver of low agreement within G2. Human rater means are nearly identical across all tasks.

A7. Radar Chart of Exact Agreement by Rater Group



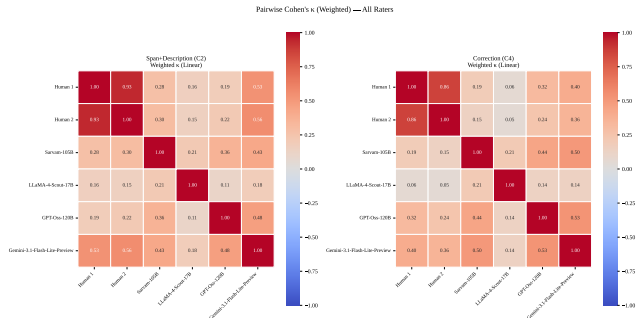
G1 (humans) has the largest polygon across all four tasks, confirming consistent dominance. G6 (Humans + GPT + Gemini) forms a noticeably larger polygon than G5 (Humans + Annotator LLMs), showing that the non-annotator LLMs are closer to humans than the self-evaluating ones. G7 has the smallest polygon, as expected.

A8. Pairwise Cohen's κ (Unweighted, All Raters)



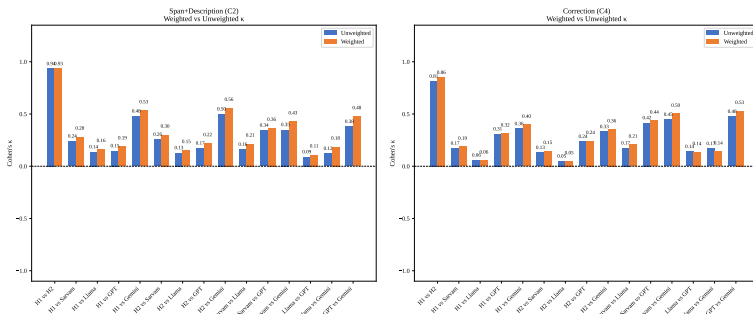
The human pair achieves $\kappa = 0.90$ to 0.94 (almost perfect). LLaMA vs. Sarvam falls to $\kappa = 0.013$ on C1 and sub-chance on C3. Gemini is the most human-aligned LLM, reaching $\kappa \approx 0.32$ to 0.48 . GPT and Gemini together achieve $\kappa = 0.53$ on C1, the strongest LLM-to-LLM pair.

A9. Pairwise Cohen's κ (Linear Weighted, C2 and C4)



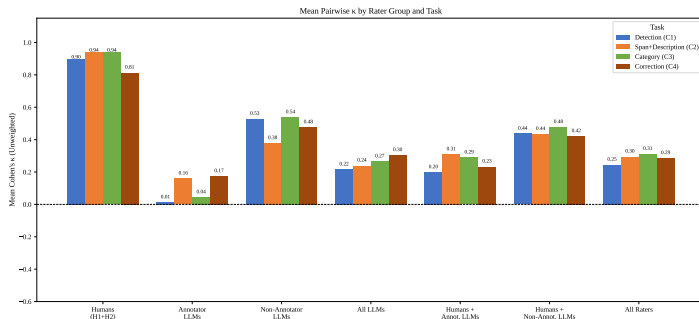
Weighted κ gives partial credit for near-misses on the ordinal scale. For the human pair it barely changes the values (the rare disagreements were already adjacent). For LLaMA-vs-Sarvam, weighted κ rises from 0.016 to about 0.21 on C2, confirming directional consistency even where exact scores differ.

A10. Weighted vs. Unweighted κ per Rater Pair (C2 and C4)



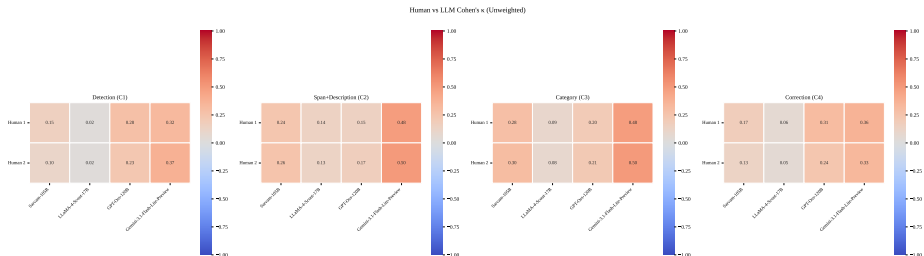
The gap between weighted and unweighted κ is largest for pairs involving LLaMA, meaning LLaMA's scores are directionally consistent with others even when not exactly matching. Human-human gaps are negligible.

A11. Mean Pairwise κ by Rater Group and Task



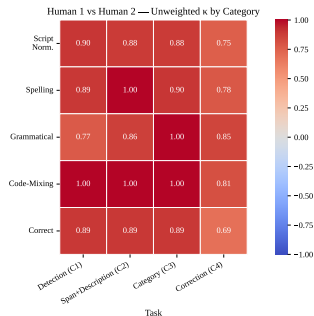
G1 leads on all tasks. G6 reaches mean $\kappa \approx 0.48$ on C3, while G5 drops to fair levels. Including the annotator LLMs in a group consistently reduces mean agreement.

A12. Human vs. LLM κ Heatmap



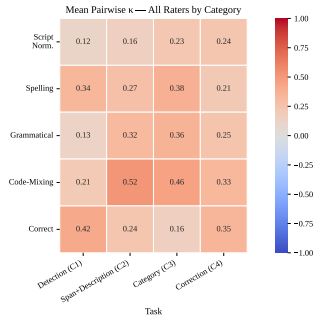
Gemini achieves the highest agreement with both humans across all tasks, peaking at $\kappa \approx 0.48$ on C3. Both Sarvam-105B and LLaMA show slight or near-zero agreement with humans on the binary tasks.

A13. Human 1 vs. Human 2 κ by Category



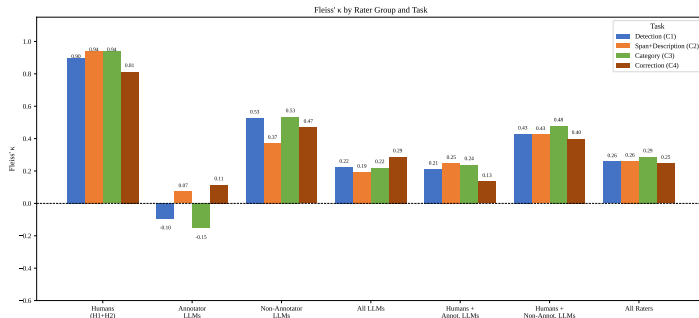
Human inter-annotator agreement is consistently high across all categories. The lowest values appear on C4 for Grammatical Error and Code-Mixing, suggesting that even expert humans find correction quality harder to judge when errors are syntactically complex.

A14. Mean Pairwise κ by Category (All Raters)



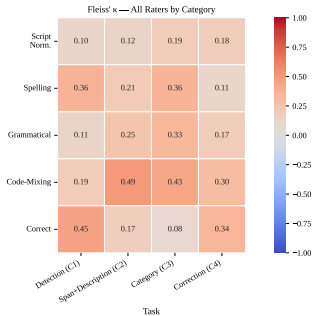
Code-Mixing achieves the highest cross-rater agreement on C1 and C3. Grammatical Error consistently shows the lowest agreement on C4, reinforcing that acceptable Odia grammatical corrections are genuinely ambiguous.

A15. Fleiss' κ by Rater Group and Task



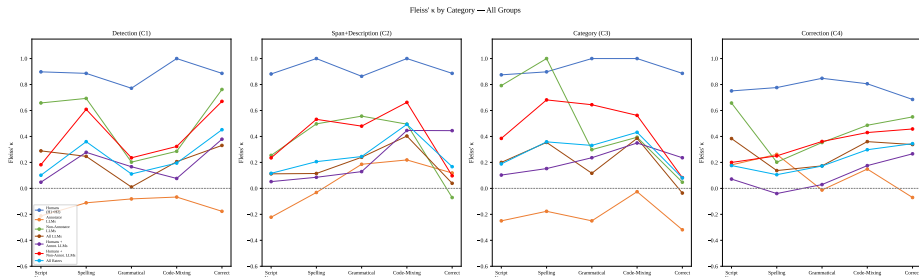
G1 achieves almost perfect $\kappa \geq 0.81$ across all tasks. G2 falls to -0.098 on C1 and -0.153 on C3, meaning Sarvam and LLaMA agree less than chance on these tasks. G6 reaches the highest multi-rater κ at 0.43 on C3.

A16. Fleiss' κ by Error Category (All Raters)



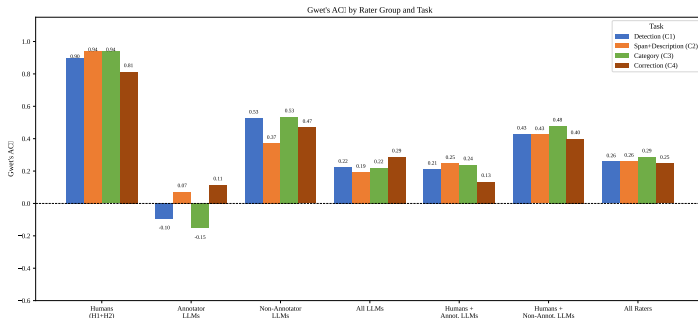
Code-Mixing consistently shows the highest all-rater κ on C1 and C3. Correct Sentence / No Errors produces moderate agreement. Grammatical Error and Script Normalization show the most variability across tasks.

A17. Fleiss' κ by Category Across All Groups



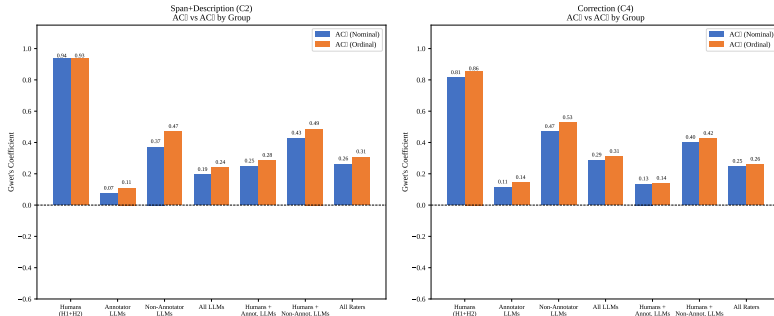
G1 maintains high κ across all categories. G2 fluctuates into negative territory for several categories on C3. G6 tracks close to G7 in shape but consistently above it, showing that adding more diverse raters reduces agreement predictably.

A18. Gwet's AC1 by Rater Group and Task



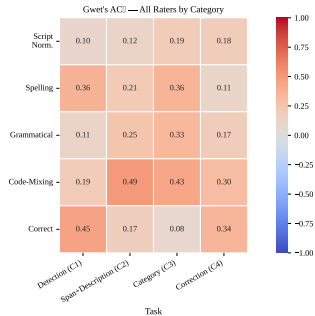
AC1 values are close to Fleiss' κ for binary tasks, confirming that the kappa paradox (prevalence-driven inflation) is not the explanation for the low G2 values. The pattern of results is therefore substantive rather than a measurement artefact.

A19. AC1 vs. AC2 for C2 and C4 (All Groups)



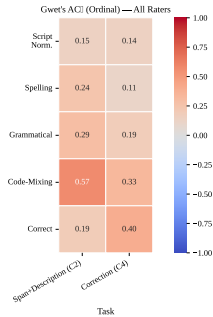
AC2 is higher than AC1 for all groups on both ordinal tasks. The gap is largest for G3 (GPT + Gemini): 0.37 vs. 0.47 on C2, indicating that these two evaluators agree on the direction of scoring even when exact scores differ.

A20. Gwet's AC1 by Category (All Raters)



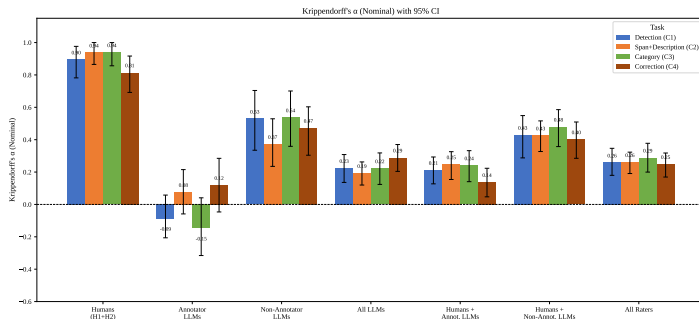
The AC1 category heatmap closely mirrors the Fleiss' κ heatmap (figure A16). Code-Mixing shows the highest values and Grammatical Error the most variability, consistent across both chance-correction frameworks.

A21. Gwet's AC2 by Category for C2 and C4 (All Raters)



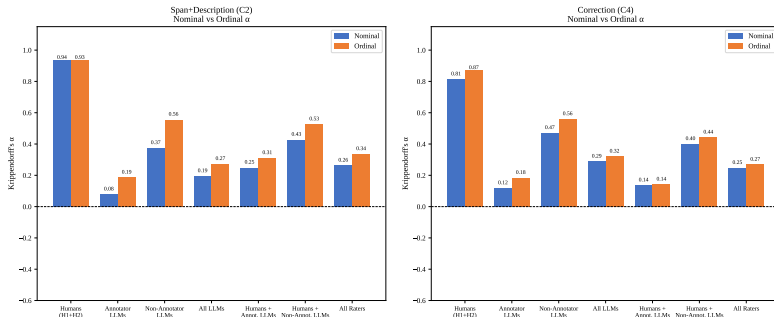
Script Normalization and Code-Mixing receive the highest ordinal agreement on correction quality (C4), consistent with these error types having relatively unambiguous correct forms. Grammatical Error shows the lowest AC2 on both C2 and C4.

A22. Krippendorff's α (Nominal) with 95% Bootstrap CI



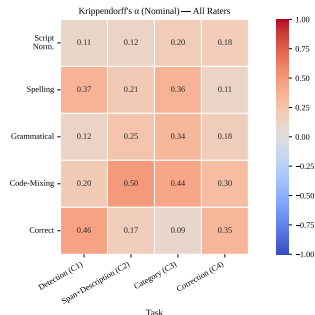
G1 exceeds $\alpha = 0.81$ on all tasks with narrow confidence intervals. G2's CIs span zero on C1 and C3, so these negative values cannot be distinguished from chance agreement. G6 reaches $\alpha = 0.43$ and 0.48 , the best multi-rater result in the study.

A23. Nominal vs. Ordinal α for C2 and C4 (All Groups)



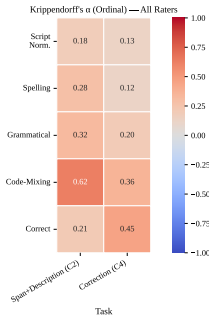
Ordinal α is consistently higher than nominal α across all groups and both tasks. The largest gain is for G3 on C2 (0.37 to 0.56), a difference of 0.19. This means that when LLMs disagree, they are rarely more than one scale step apart, so nominal measurement understates their actual consistency.

A24. Krippendorff's α (Nominal) by Category (All Raters)



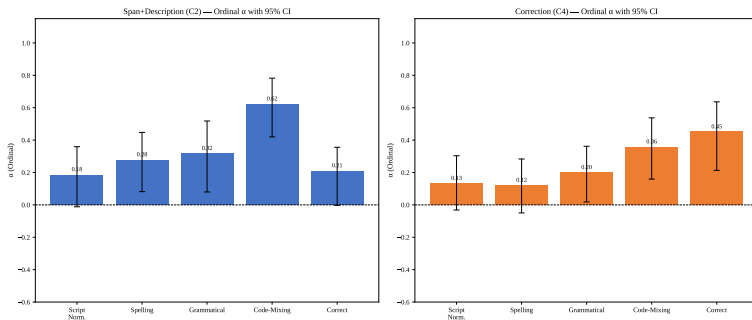
Code-Mixing is the most reliably evaluated category across both nominal and ordinal measures. Correct Sentence / No Errors shows the widest disagreement range, possibly because evaluators differ on whether a truly error-free sentence warrants any annotation at all.

A25. Krippendorff's α (Ordinal) by Category for C2 and C4



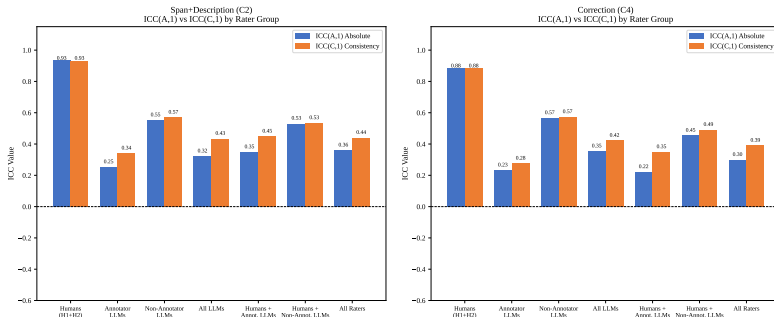
Script Normalization shows moderate ordinal α on C2, suggesting that while the error type is identifiable, describing the exact normalisation rule introduces evaluator variability. Grammatical Error has the lowest ordinal α on C4.

A26. Ordinal α with Bootstrap CI by Category (All Raters)



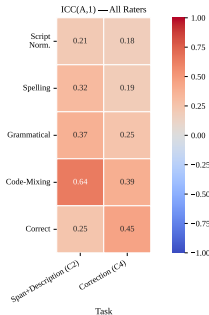
With only 20 items per category, the CIs are wide; category-level results are therefore exploratory rather than confirmatory. Code-Mixing shows the tightest CI and highest point estimates. Correct Sentence / No Errors shows the widest uncertainty on both tasks.

A27. ICC(A,1) and ICC(C,1) by Rater Group (C2 and C4)



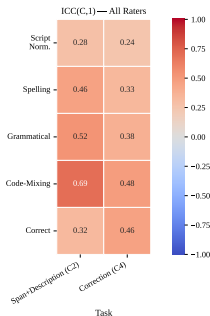
G1 achieves excellent $ICC(A,1) = ICC(C,1) = 0.933$ on C2 and good reliability on C4 (0.884). G2 shows poor to moderate reliability with a visible $ICC(C,1)$ -to- $ICC(A,1)$ gap, indicating consistent ranking but different absolute thresholds. G6 reaches moderate reliability (0.53 on C2).

A28. ICC(A,1) by Category (All Raters)



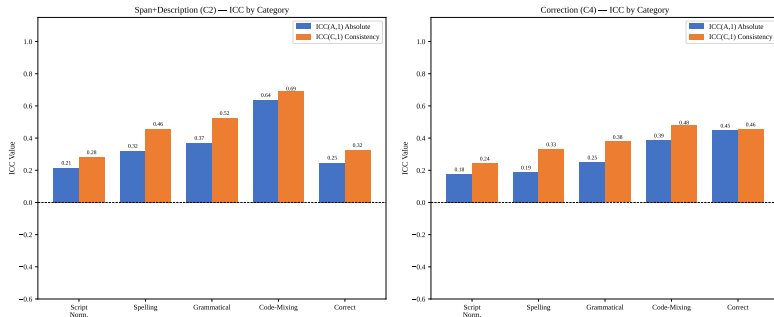
Code-Mixing and Correct Sentence / No Errors show the highest ICC on C2. Grammatical Error consistently shows the lowest values on C4, confirming that correction quality for complex syntactic errors is the hardest dimension to evaluate reliably.

A29. ICC(C,1) by Category (All Raters)



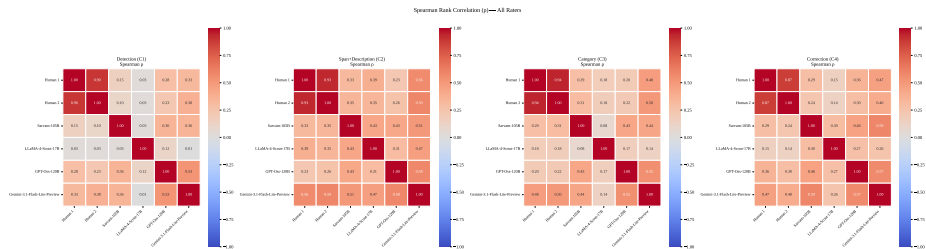
ICC(C,1) values are uniformly higher than the corresponding ICC(A,1) values in figure A28, confirming that directional rank agreement exceeds absolute score agreement across all categories. The pattern of relative difficulty across categories is the same in both maps.

A30. ICC(A,1) vs. ICC(C,1) by Category (All Raters)



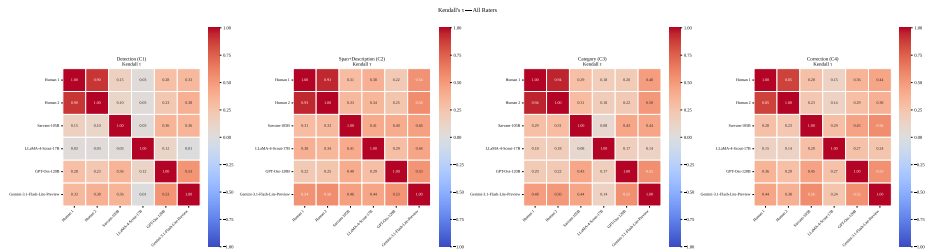
The gap between ICC(A,1) and ICC(C,1) measures systematic evaluator bias per category. The gap is largest for Script Normalization on C2, suggesting evaluators agree on which annotation is better but differ in how strictly they calibrate the 0/1/2 boundary for span quality.

A31. Spearman's ρ : All Evaluator Pairs, All Tasks



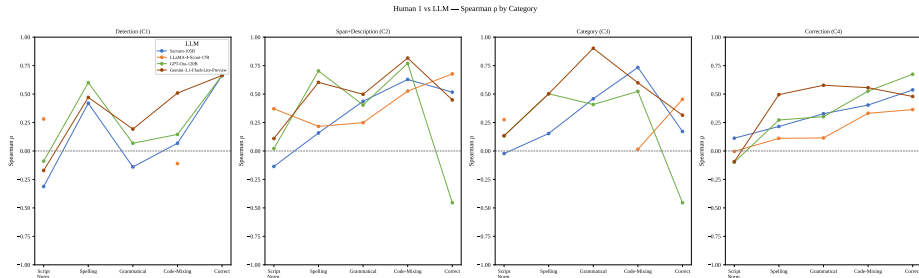
The human pair shows $\rho = 0.90$ to 0.94 across all tasks (all $p < 0.001$). Gemini correlates most strongly with humans among the LLMs ($\rho \approx 0.37$ on C1, ≈ 0.56 on C2). LLaMA vs. Sarvam shows $\rho = 0.025$ on C1, statistically indistinguishable from zero.

A32. Kendall's τ : All Evaluator Pairs, All Tasks



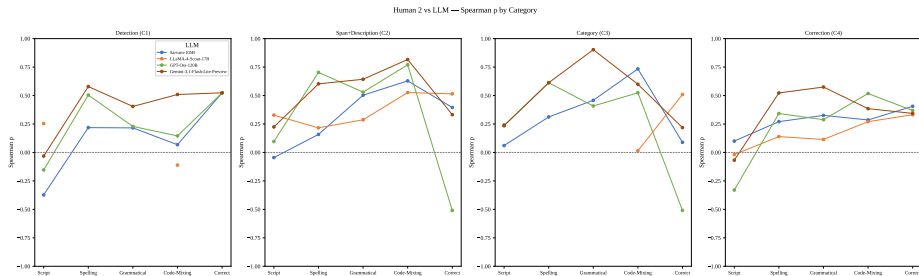
Kendall's τ values are uniformly slightly lower than the corresponding Spearman values, as expected. The relative ordering of rater pairs is preserved, and the convergence of both rank-correlation measures strengthens confidence in the correlation findings.

A33. Spearman's ρ : Human 1 vs. LLMs by Category



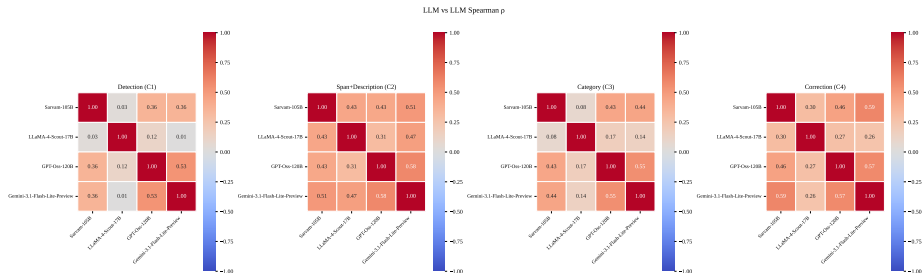
Gemini consistently achieves the highest ρ with Human 1 across all tasks and categories. Code-Mixing and Script Normalization show the highest human-LLM correlations. Grammatical Error shows the lowest, particularly on C4. LLaMA shows near-zero or negative correlations on C1 for several categories.

A34. Spearman's ρ : Human 2 vs. LLMs by Category



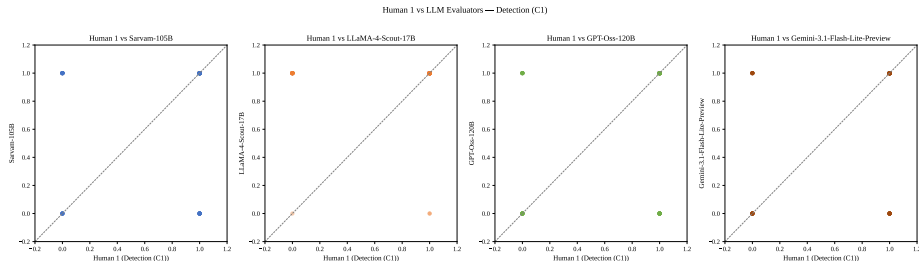
The pattern mirrors figure A33: Gemini is the most human-aligned LLM across all tasks for Human 2 as well. The consistency across both humans confirms that Gemini's alignment is not an artefact of one evaluator's particular scoring tendencies.

A35. Spearman's ρ : LLM Evaluators Only



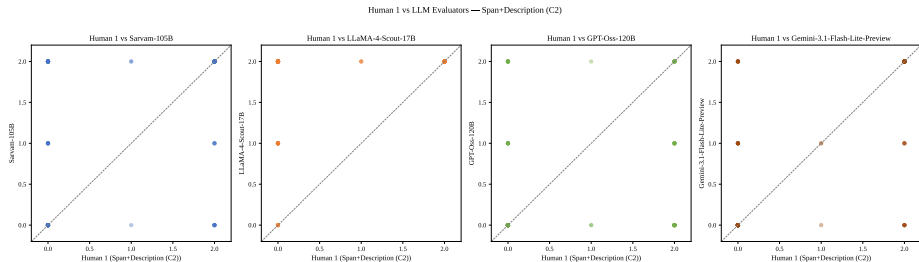
GPT-Oss-120B and Gemini show the strongest LLM-to-LLM correlation ($\rho = 0.53$ to 0.58), suggesting they have converged on similar evaluation heuristics. LLaMA's near-zero correlation with Sarvam on C1 ($\rho = 0.025$) is confirmed here as a structural feature of its idiosyncratic error detection judgements.

A36. Scatter Plots: Human 1 vs. LLMs for C1



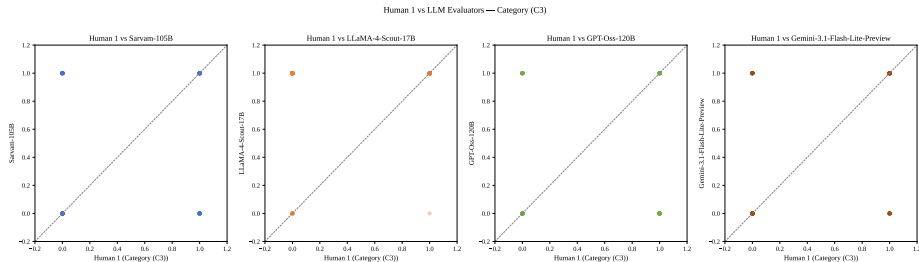
LLaMA shows the most off-diagonal scatter against Human 1 on C1, with many items where LLaMA scores 1 and the human scores 0 (false positives). Gemini's scatter is more concentrated on the diagonal.

A37. Scatter Plots: Human 1 vs. LLMs for C2



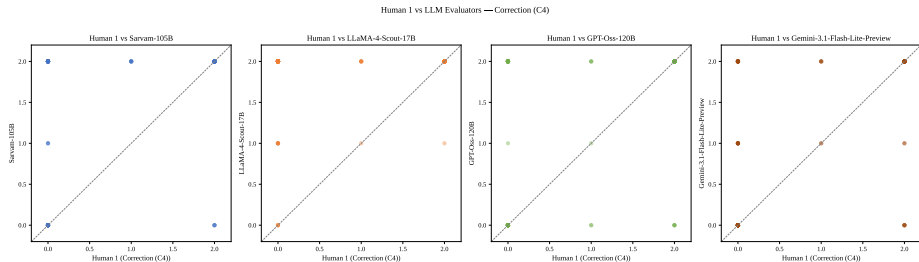
LLaMA's tendency to assign 2 where Human 1 assigns 0 or 1 is visible as a cluster of points in the upper-left quadrant of LLaMA's panel. Gemini shows the tightest diagonal clustering of any LLM on C2.

A38. Scatter Plots: Human 1 vs. LLMs for C3



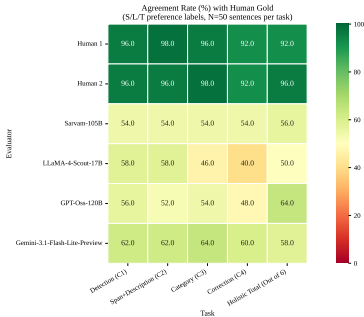
Category assignment (C3) is binary, so scatter plots reduce to four quadrants. LLaMA has the most observations in the off-diagonal quadrants (human says 0, LLaMA says 1, or vice versa), consistent with its sub-chance κ on this task.

A39. Scatter Plots: Human 1 vs. LLMs for C4



On C4 (correction quality), LLaMA's leniency bias appears as a strong upward shift: many items scored 1 or 2 by LLaMA are scored 0 by Human 1. Sarvam-105B is closer to the diagonal but still elevated relative to the human standard.

A40. Agreement Rate (%) with Human Gold per Evaluator



Human raters reach 92–98% agreement with the gold S/L/T label. Gemini leads LLM evaluators at 60–64%. LLaMA drops to 40% on C4, the lowest single cell in the table, consistent with its strong self-preference bias on correction quality.

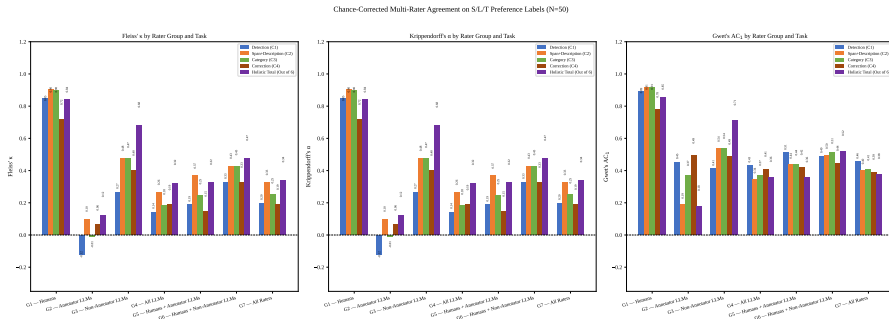


A41. Agreement Rate (%) by Evaluator, Category and Task



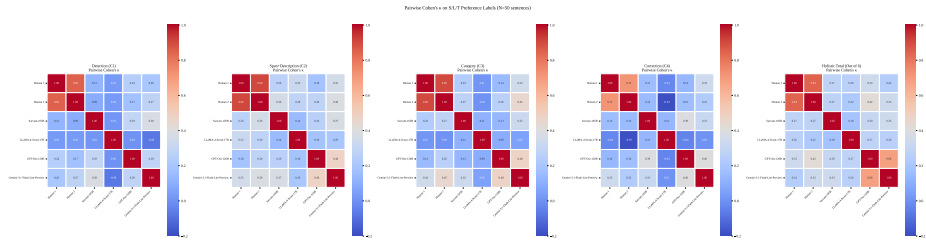
Human evaluators achieve near-perfect agreement rates on Grammatical Error and Script Normalization across all tasks. LLaMA shows the most pronounced category-level variation, with markedly low agreement on Spelling and Code-Mixing, consistent with its optimism bias being strongest there.

A42. Chance-Corrected Multi-Rater Agreement (Fleiss' κ , Krippendorff's α , Gwet's AC₁)



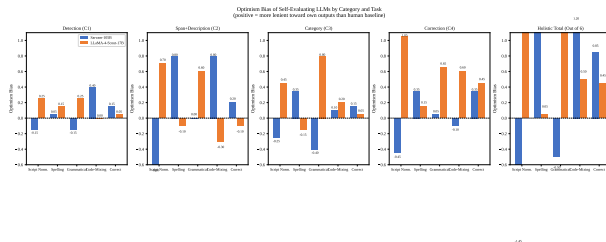
All three metrics agree in direction. G1 (humans) consistently exceeds 0.72 across all tasks. G2 (annotator LLMs) records Fleiss' $\kappa < 0$ on two tasks, confirming sub-chance agreement between Sarvam and LLaMA. G3 (non-annotator LLMs) outperforms G2 on every cell, demonstrating that removing the conflict-of-interest substantially improves group agreement.

A43. Pairwise Cohen's κ on S/L/T Preference Labels



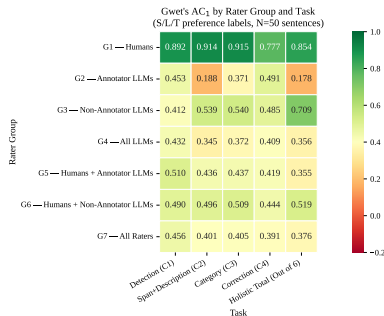
The human pair achieves $\kappa = 0.85\text{--}0.90$ (almost perfect). LLaMA vs. Sarvam falls to $\kappa = -0.12$ on C1, the only negative pairwise cell in the matrix, confirming systematic opposition between the two self-evaluating annotators. Gemini is the most human-aligned LLM pairwise evaluator.

A44. Optimism Bias of Self-Evaluating LLMs by Category



Optimism bias is the self-evaluator's score margin minus the human gold margin. LLaMA shows the strongest self-preference across all tasks and categories: +0.58 on C4, +0.27 on C3. Sarvam-105B is near zero on C1 and C3, indicating substantially better calibration, consistent with Panickssery, Bowman, & Feng (2024).

A45. Gwet's AC₁ by Rater Group and Task



G1 ranges from 0.78 (C4) to 0.91 (C2–C3), confirming robust human consensus. G2 drops to 0.19 on C2, the weakest group-task cell in the study. G3 uniformly exceeds G2, and G6 exceeds G5 on every task, reinforcing that non-annotator LLMs are more reliable preference evaluators than the self-evaluating annotator LLMs.

A46. Sentence-Level Score Variance and Delta

Section	Category	Sent. ID	Source Sentence	H1	H2	Sarvam	LLaMA	GPT	Gemini	Delta
Most agreed	Script Norm.	SN_03	ମୋହନ ଭାଗଓବ୍ କହିଛନ୍ତି, "ଏବେ ହେବ ରାମଙ୍କ କାମା"	S:0 L:0	S:0 L:0	S:6 L:6	S:6 L:6	S:5 L:6	S:5 L:3	1
Most disagreed	Script Norm.	SN_02	କୃଷି କାର୍ଯ୍ୟ ଯେପରି ଠିକ୍ ସମୟରେ ହୋଇପାରିବ, ସେଥି ନିମନ୍ତେ ଉଭୟ ସ୍ୱଳ୍ପ ମିଆଦି ଓ ଚାନ୍ଦ ମିଆଦି ଯୋଜନା ପ୍ରସ୍ତୁତ କରିବା ଉପରେ ଗୁରୁତ୍ୱରୋପ କରାଯାଇଥିଲା।	S:6 L:0	S:6 L:0	S:2 L:3	S:6 L:5	S:0 L:4	S:1 L:4	5.27
Most agreed	Spelling	ST_08	ଶିକ୍ଷାର ମହତ୍ତ୍ୱ ଅଗଣା	S:3 L:2	S:3 L:2	S:6 L:4	S:6 L:6	S:6 L:5	S:4 L:3	0.67
Most disagreed	Spelling	ST_07	ସେହି ସମୟରେ ଏପରି ବେଦ ଏବଂ ପୁରାଣ ଲେଖାଯାଇଥିଲା, ଯାହା ଅତି ବି ଜ୍ଞାନର ଶ୍ରୋତ ଅଟେ।	S:1 L:2	S:1 L:2	S:6 L:0	S:6 L:4	S:5 L:0	S:2 L:0	3.53
Most agreed	Grammatical	GT_06	ମୁଁ ପ୍ରଥମେ ମୋର କଲେଜ ବନ୍ଧୁମାନଙ୍କ ସହିତ ବାହାରି ପଡ଼ିଗଲା।	S:1 L:1	S:1 L:1	S:1 L:2	S:5 L:4	S:2 L:0	S:0 L:0	1.2
Most disagreed	Grammatical	GT_02	ଏହାପ୍ରାୟ ଶିକ୍ଷଣ ଦ୍ୱାରରେ ଯଥେଷ୍ଟ ସୁଧାର ଆସିଛନ୍ତି।	S:6 L:0	S:6 L:0	S:6 L:6	S:6 L:6	S:6 L:3	S:6 L:0	3.4
Most agreed	Code-Mixing	WL_03	Christiano ପିଲାଦିନରୁ ବହୁତ ଗରିବ ଥିଲେ।	S:6 L:5	S:6 L:5	S:6 L:6	S:6 L:6	S:6 L:6	S:6 L:6	0.53
Most disagreed	Code-Mixing	WL_09	ଇତିହାସରେ ଅନେକ କାହାଣୀର ପ୍ରମାଣ ଆଛି ଏବଂ ତାହା ମଧ୍ୟ ବିଭିନ୍ନ ଚକ୍ରମାନୁସାରେ।	S:1 L:5	S:1 L:5	S:6 L:2	S:6 L:6	S:2 L:5	S:6 L:6	3.67
Most agreed	Correct	CS_01	ଶିକ୍ଷା କ'ଣ?	S:6 L:6	S:6 L:6	S:5 L:5	S:6 L:6	S:3 L:3	S:6 L:6	0
Most disagreed	Correct	CS_07	ଏହା ବ୍ୟତୀତ, ଏହା ମଧ୍ୟ ଦେଖାଯାଇଛି ଯେ ସାଧାରଣ ଲୋକମାନେ ଏ ବିଷୟରେ ସେତେ ଜାଗରୁକ ନାହାଁନ୍ତି।	S:0 L:0	S:0 L:0	S:6 L:1	S:2 L:6	S:0 L:6	S:0 L:6	5.13

This table details the raw score distributions (Sarvam vs. LLaMA) and the resulting maximum delta (variance) across raters. Sentences with minimal delta, such as CS_01 (Correct Sentence, Delta = 0), demonstrate calibrated scoring where all raters reliably assign tied max scores. In extreme contrast, CS_07 produces a massive delta of 5.13. For this sentence, humans assign absolute penalties (S:0 L:0), whereas LLMs exhibit extreme polarity: Sarvam scores itself highly (S:6 L:1) and the remaining models overwhelmingly favor LLaMA (L:6).



A47. Sentence-Level System Preference Agreement

Section	Category	Sent. ID	Source Sentence	H 1	H 2	Sarvam	LLaMA	GPT	Gemini	Agreement
Most agreed	Script Norm.	SN_08	ବିଭିନ୍ନ ଭାବରେ ମୂର୍ତ୍ତିପୂଜାର ଆଦର କରିଥିଲେ ମଧ୍ୟ ମୂଳ ଚତୁଷ୍ଟୁ ବାଦ ଦେବା ସମ୍ଭବ ହୋଇନାହିଁ।	T	T	T	T	T	T	0.767
Most disagreed	Script Norm.	SN_02	କୃଷି କାର୍ଯ୍ୟ ଯେପରି ଠିକ୍ ସମୟରେ ହୋଇପାରିବ, ସେଥି ନିମନ୍ତେ ଭଲ ସ୍ଥଳ ମିଆଁ ଓ ଦାଢ଼ ମିଆଁ ଯୋଜନା ପ୍ରସ୍ତୁତ କରିବା ଉପରେ ଗୁରୁତ୍ୱାବେଶ କରାଯାଇଥିଲା।	S	S	T	T	L	L	0.267
Most agreed	Spelling	ST_06	ମଣିଷ ପ୍ରତ୍ୟେକ ପର୍ଯ୍ୟୟରୁ ଶାଖେ।	T	T	T	T	T	T	0.833
Most disagreed	Spelling	ST_07	ଯେହି ସମୟରେ ଏପରି ବେଦ ଏବଂ ପୁରାଣ ଲେଖାଯାଇଥିଲା, ଯାହା ଆଜି ବି ଜ୍ଞାନର ଶ୍ରୋତ ଅଟେ।	T	T	S	T	S	T	0.367
Most agreed	Grammatical	GT_01	ଆଦିମାନଙ୍କ ମଧ୍ୟ ଶିଖି ଶିଖି ଏହି ସୁଦ୍ଧାରେ ପଡ଼ିଥିବା ଯେ ଖାଦ୍ୟ ରାନ୍ଧିବି ଖାଇବାକୁ ଲାଗିଥିଲେ।	S	S	S	T	T	S	0.7
Most disagreed	Grammatical	GT_02	ଏହାଦ୍ୱାରା ଶିକ୍ଷଣ ଚାନ୍ଦରେ ଯଥେଷ୍ଟ ସୁଧାର ଆସିଛନ୍ତି।	S	S	T	T	T	S	0.433
Most agreed	Code-Mixing	WL_03	Christiano ଦିଲ୍ଲୀଦିଲ୍ଲୀ ବହୁତ ଭାରିବ ଥିଲେ।	T	T	T	T	T	T	0.867
Most disagreed	Code-Mixing	WL_01	ଆଜିକାଲି 90 ପ୍ରତିଶତରୁ ଅଧିକ ଲୋକ ସାମାଜିକ ଗଣମାଧ୍ୟମ ସହିତ ଜଡ଼ିତ।	T	T	S	T	S	S	0.417
Most agreed	Correct	CS_01	ଶିକ୍ଷା କ'ଣ?	T	T	T	T	T	T	1
Most disagreed	Correct	CS_07	ଏହା ବ୍ୟତୀତ, ଏହା ମଧ୍ୟ ଦେଖାଯାଇଛି ଯେ ସାଧାରଣ ଲୋକମାନେ ଏ ବିଷୟରେ ସେତେ ଜାଗରୁକ ନାହାଁନ୍ତି।	T	T	S	L	L	L	0.267

This table highlights specific sentences driving the highest and lowest inter-rater preference agreement. For highly agreed sentences like CS_01 (Correct) and WL_03 (Code-Mixing), all LLM evaluators perfectly or near-perfectly align with the human consensus of a 'Tie' (T). Conversely, on highly disagreed sentences like CS_07 (Correct) and SN_02 (Script Normalization), evaluators fragment completely. For instance, on CS_07, human raters assign a Tie, but Sarvam votes for itself ('S') while LLaMA, GPT, and Gemini all vote for LLaMA ('L'), dropping agreement to 0.267.



A48. Unweighted Pairwise Cohen's κ (All Tasks)

Pair	C1 Detection	C2 Span+Desc.	C3 Category	C4 Correction
H1 vs H2	0.90	0.94	0.94	0.81
H1 vs Sarvam	0.15	0.24	0.28	0.17
H1 vs LLaMA	0.02	0.14	0.09	0.06
H1 vs GPT	0.28	0.15	0.20	0.31
H1 vs Gemini	0.32	0.48	0.48	0.36
H2 vs Sarvam	0.10	0.26	0.30	0.13
H2 vs LLaMA	0.02	0.13	0.08	0.05
H2 vs GPT	0.23	0.17	0.21	0.24
H2 vs Gemini	0.37	0.50	0.50	0.33
Sarvam vs LLaMA	0.01	0.16	0.04	0.17
Sarvam vs GPT	0.35	0.34	0.42	0.42
Sarvam vs Gemini	0.35	0.35	0.44	0.45
LLaMA vs GPT	0.05	0.09	0.08	0.14
LLaMA vs Gemini	0.00	0.12	0.08	0.17
GPT vs Gemini	0.53	0.38	0.54	0.48

A49. Linear Weighted Cohen's κ (C2 and C4)

Pair	C2 Span+Description	C4 Correction
H1 vs H2	0.93	0.86
H1 vs Sarvam	0.28	0.19
H1 vs LLaMA	0.16	0.06
H1 vs GPT	0.19	0.32
H1 vs Gemini	0.53	0.40
H2 vs Sarvam	0.30	0.15
H2 vs LLaMA	0.15	0.05
H2 vs GPT	0.22	0.24
H2 vs Gemini	0.56	0.36
Sarvam vs LLaMA	0.21	0.21
Sarvam vs GPT	0.36	0.44
Sarvam vs Gemini	0.43	0.50
LLaMA vs GPT	0.11	0.14
LLaMA vs Gemini	0.18	0.14
GPT vs Gemini	0.48	0.53

A50. Reliability Gain: Weighted vs. Unweighted Gap

Pair	C2 Gap	C4 Gap
H1 vs H2	-0.01	+0.05
H1 vs Sarvam	+0.04	+0.02
H1 vs LLaMA	+0.02	0.00
H1 vs GPT	+0.04	+0.01
H1 vs Gemini	+0.05	+0.04
H2 vs Sarvam	+0.04	+0.02
H2 vs LLaMA	+0.02	0.00
H2 vs GPT	+0.05	0.00
H2 vs Gemini	+0.06	+0.03
Sarvam vs LLaMA	+0.05	+0.04
Sarvam vs GPT	+0.02	+0.02
Sarvam vs Gemini	+0.08	+0.05
LLaMA vs GPT	+0.02	0.00
LLaMA vs Gemini	+0.06	-0.03
GPT vs Gemini	+0.10	+0.05

A51. Raw Exact Agreement by Rater Group

Task	G1	G2	G3	G4	G5	G6	G7
C1 Detection	96%	72%	79%	48%	55%	59%	41%
C2 Span+Desc.	97%	51%	64%	18%	28%	48%	13%
C3 Category	97%	53%	77%	32%	33%	55%	25%
C4 Correction	91%	68%	70%	37%	24%	49%	20%

A52. Fleiss' κ by Rater Group

Task	G1	G2	G3	G4	G5	G6	G7
C1 Detection	0.90	-0.10	0.53	0.22	0.21	0.43	0.26
C2 Span+Desc.	0.94	0.07	0.37	0.19	0.25	0.43	0.26
C3 Category	0.94	-0.15	0.53	0.22	0.24	0.48	0.29
C4 Correction	0.81	0.11	0.47	0.29	0.13	0.40	0.25

A53. Gwet's AC_1 (Nominal Agreement)

Task	G1	G2	G3	G4	G5	G6	G7
C1 Detection	0.90	−0.10	0.53	0.22	0.21	0.43	0.26
C2 Span+Desc.	0.94	0.07	0.37	0.19	0.25	0.43	0.26
C3 Category	0.94	−0.15	0.53	0.22	0.24	0.48	0.29
C4 Correction	0.81	0.11	0.47	0.29	0.13	0.40	0.25

A54. Krippendorff's α (Nominal)

Task	G1	G2	G3	G4	G5	G6	G7
C1 Detection	0.90	-0.09	0.53	0.23	0.21	0.43	0.26
C2 Span+Desc.	0.94	0.08	0.37	0.19	0.25	0.43	0.26
C3 Category	0.94	-0.15	0.54	0.22	0.24	0.48	0.29
C4 Correction	0.81	0.12	0.47	0.29	0.14	0.40	0.25

A55. Ordinal Reliability: Krippendorff's α and Gwet's AC_2

Method/Task	G1	G2	G3	G4	G5	G6	G7
α (Ordinal)							
C2 Span+Desc.	0.93	0.19	0.56	0.27	0.31	0.53	0.34
C4 Correction	0.87	0.18	0.56	0.32	0.14	0.44	0.27
AC_2 (Ordinal)							
C2 Span+Desc.	0.93	0.11	0.47	0.24	0.28	0.49	0.31
C4 Correction	0.86	0.14	0.53	0.31	0.14	0.42	0.26

A56. Intraclass Correlation Coefficient (ICC)

Metric/Task	G1	G2	G3	G4	G5	G6	G7
ICC(A,I)	0.93	0.25	0.55	0.32	0.35	0.53	0.36
ICC(C,I)	0.93	0.34	0.57	0.43	0.45	0.53	0.44
Gap (C,I – A,I)	0.00	0.09	0.02	0.11	0.10	0.00	0.08

A57. Gwet's AC_1 on System Preference (S/L/T)

Task	G1	G2	G3	G4	G5	G6	G7
C1 Detection	0.892	0.453	0.412	0.432	0.510	0.490	0.456
C2 Span+Desc.	0.914	0.188	0.539	0.345	0.436	0.496	0.401
C3 Category	0.915	0.371	0.540	0.372	0.437	0.509	0.405
C4 Correction	0.777	0.491	0.485	0.409	0.419	0.444	0.391

A58. Spearman's ρ Rank Correlation (All Tasks)

Pair	C1 Detection	C2 Span+Desc.	C3 Category	C4 Correction
H1 vs H2	0.90	0.93	0.94	0.87
H1 vs Sarvam	0.15	0.33	0.29	0.29
H1 vs LLaMA	0.03	0.39	0.18	0.15
H1 vs GPT	0.28	0.23	0.20	0.36
H1 vs Gemini	0.33	0.56	0.48	0.47
H2 vs Sarvam	0.10	0.35	0.31	0.24
H2 vs LLaMA	0.03	0.35	0.18	0.14
H2 vs GPT	0.23	0.26	0.22	0.30
H2 vs Gemini	0.38	0.59	0.50	0.40
Sarvam vs LLaMA	0.03	0.43	0.08	0.30
Sarvam vs GPT	0.36	0.43	0.43	0.46
Sarvam vs Gemini	0.36	0.51	0.44	0.59
LLaMA vs GPT	0.12	0.31	0.17	0.27
LLaMA vs Gemini	0.01	0.47	0.14	0.26
GPT vs Gemini	0.53	0.58	0.55	0.57

A59. Kendall's τ Rank Correlation (All Tasks)

Pair	C1 Detection	C2 Span+Desc.	C3 Category	C4 Correction
H1 vs H2	0.90	0.93	0.94	0.85
H1 vs Sarvam	0.15	0.31	0.29	0.28
H1 vs LLaMA	0.03	0.38	0.18	0.15
H1 vs GPT	0.28	0.22	0.20	0.36
H1 vs Gemini	0.33	0.54	0.48	0.44
H2 vs Sarvam	0.10	0.33	0.31	0.23
H2 vs LLaMA	0.03	0.34	0.18	0.14
H2 vs GPT	0.23	0.25	0.22	0.29
H2 vs Gemini	0.38	0.56	0.50	0.38
Sarvam vs LLaMA	0.03	0.41	0.08	0.29
Sarvam vs GPT	0.36	0.40	0.43	0.45
Sarvam vs Gemini	0.36	0.46	0.44	0.56
LLaMA vs GPT	0.12	0.29	0.17	0.27
LLaMA vs Gemini	0.01	0.44	0.14	0.24
GPT vs Gemini	0.53	0.53	0.55	0.55

A60. Methodological Divergence: ρ vs. τ Gap

Pair	C2 Gap ($\rho - \tau$)	C4 Gap ($\rho - \tau$)
H1 vs H2	0.00	+0.02
H1 vs Gemini	+0.02	+0.03
H2 vs Gemini	+0.03	+0.02
Sarvam vs LLaMA	+0.02	+0.01
Sarvam vs Gemini	+0.05	+0.03
LLaMA vs Gemini	+0.03	+0.02
GPT vs Gemini	+0.05	+0.02